Abhisek Ghosal

Department of Chemistry, Northwestern University
2145 Sheridan Road Evanston, IL 60208-3113
Phone: (+1) 708-971-4709 (+91) 954-784-2600
Email: · abhisek.ghosal@northwestern.edu · abhisek.ghoshal1990@gmail.com

Research Experience

Northwestern University, Department of Chemistry — Evanston, IL, USA
Postdoctoral Research Scholar, Prof. George C. Schatz Group — Sep. 2023-Present

- **Theoretical characterization of autoionizing excited states and their role in chemical dynamics.**
  - Developing non-Hermitian Hamiltonian based quantum chemical methods for Feshbach and shape-type of resonances.
  - Computing the rate of collision-induced associative ionization processes in plasma formation.
  - Modelling the photodynamics of Green Fluorescent Protein involving autoionizing excited-state surfaces.
- **Electron–electron scattering–induced relaxation (EESR) of plasmonic hot carriers (HCs) from a many-body theoretical framework and its role in photocatalysis.**
  - Developing an efficient implementation of the self-energy at the GW/RPA level using multipole expansion for large plasmonic systems.
  - Modeling tractable simulations of quantum plasmonic processes relevant to photocatalysis, including EESR of plasmonic HCs.

TIFR Mumbai, Department of Chemical Sciences — Mumbai, MH, India
Visiting postdoctoral fellow, Dr. Vamsee K. Voora Group — May. 2021-Aug. 2023

- **Theoretical characterization of negative-ion resonances using complex density functional theory**
  - Developing efficient one-particle nonlocal exchange–correlation potentials by combining exact-exchange with random phase approximation (RPA) correlation.
  - Accessing negative-ion resonances by perturbing real, Hermitian nonlocal exchange–correlation (XC) potentials with complex absorbing local potentials.
- **Modeling the optical gap of F-center defects using many-body theory**
  - Developing efficient one-particle nonlocal exchange–correlation (XC) potentials by combining exact exchange with single-pole RPA correlation.
  - Accessing bulk and surface F-center defects by combining RPA based potentials with periodic electrostatic embedding.

IISER Kolkata, Department of Chemical Sciences — Mohanpur, WB, India
Graduate Researcher, Prof. Amlan K. Roy Group — Oct. 2014-Apr. 2021

- **Development of real-space density functional theory (DFT) for many-electron systems: electronic structure and dynamics**
  - Development of DFT methods within a Cartesian coordinate grid (CCG) framework.

- Methodological development of efficient Hartree–Fock exchange energy densities for hyper exchange–correlation (XC) functionals.
- Self-consistent and systematic optimization of range-separated hybrid functionals from *first principles.*
- Real-time time-dependent DFT in CCG for investigating nonlinear optical properties of many-electron systems.
- Modelling Charge-transfer excited states through Becke's exciton model within a Hybrid generalised Kohn-Sham framework.

Education

Indian Institute of Science Education and Research, Kolkata — Mohanpur, WB
PhD., Theoretical Chemistry — 2014 - 2021

- CSIR/UGC-SRF Awardee Dec-2016
- CSIR/UGC-JRF Awardee Dec-2014
- **Research Advisor:** Prof. Amlan K Roy

Presidency University — Kolkata
M.Sc., Physical Chemistry — 2011-2013

- Degree Obtained June 2014
- **M.Sc. Thesis Advisor:**
  Prof. Sanjib Ghosh, former Vice-Chancellor of Presidency college

Presidency College, University of Calcutta — Kolkata
B.Sc., Chemistry(Honours), Physics and Mathematics — 2009-2011

Bhanderhati B.M. Institution, WBSHSE — Hooghly
High School graduated, — 2009

Fellowships/Grants/Awards/Honors

- **Early Career Researcher Prizes, Molecular Physics** — July 2022
  (*An International Journal at the Interface Between Chemistry and Physics*)
- **International Travel Grant, DST-SERB, Gov. of India** — *Aug. 2019*
- **Senior Research Fellowship, CSIR/UGC SRF, Gov. of India** — *Dec. 2017*
- **Junior Research Fellowship, CSIR/UGC JRF, Gov. of India** — *Dec. 2014*
- **Best Poster, Departmental Day of Chemical Sciences, IISER Kolkata** — *Dec. 2015*
- **Secured a Leading Position, JAM-IITs, Gov. od India** — *2011*
  (Joint admission in Masters - Indian Institute of Technologies)

Teaching Experience

IISER Kolkata, Department of Chemical Sciences — Mohanpur, WB
Teaching Assistant — Autumn Semester, 2018

- CH4102: Molecular Structure and Symmetry

IISER Kolkata, Department of Chemical Sciences Mohanpur, WB
Teaching Assistant Autumn Semester, 2015

- CH4102: Molecular Structure and Symmetry

Mentoring Experience

- **Mr. Kishalay Mahato** BS-MS 2014, IISER Kolkata
  (*University of Maryland, USA)*
- **Mr. Tarun Gupta** *BS-MS 2015, IISER Kolkata*
  (University of Warsaw, Poland)
- **Mr. Raj Roy** *JRF 2020, IISER Kolkata*
  (New York University, Shanghai)
- **Mr. Sayan Adhikari** *BS-MS 2021, IISER Berhampur*
  (University of California, Merced)

Presentations

- **Symposiums**
  - Abhisek Ghosal and George C. Schatz, CMQT Symposium, Northwestern University, (May 1-2, 2025).
  - Abhisek Ghosal and George C. Schatz, IIN symposium, Northwestern University, (October 10, 2024).
- **Invited Talks**
  - *Complex Density Functional Theory : A Non-Hermitian approach to Negative ion Resonances*, CASUS Institute Seminar, Helmholtz-Zentrum Dresden-Rossendorf e.V. (HZDR), October 8, (2024).
  - *Theoretical basis and accurate approximations for modelling negative ion resonances*, Prof. William Glover Group, NYU Shanghai, June 26, (2024).
  - *Taming negative ion resonances using non-local density functionals*, Prof. Stefano Sanvito Group, Trinity College, Dublin, November 17, (2023).
  - *Density functional theory of many-electron systems in Cartesian grid*, on GIAN course of MHRD, Gov. of India organized by IIT-Bhubaneswar. (8-18 July, 2016).
- **Posters**
  - Abhisek Ghosal and Amlan K Roy, *A self-consistent systematic optimization of range-separated hybrid functionals from first principles*, Virtual conference on recent development of electronic structure method, Department of Physics, UC Merced. (2-4 June, 2020).
  - Abhisek Ghosal and Amlan K Roy, *A self-consistent systematic optimization of range-separated hybrid functionals from first principles*, 2020 New horizons in density functional theory Faraday discussions, Royal Society of Chemistry, UK. (2-4 September, 2020).
  - Abhisek Ghosal and Amlan K Roy, *Real-time time-dependent density functional theory*

of many-electron systems in Cartesian grid, 2019 TDDFT School & Workshop: Excited states and dynamics, Rutgers University-Newark. (5-14 August, 2019).
  - Abhisek Ghosal and Amlan K Roy, *A pseudopotential Cartesian grid based KS-DFT for many-electron systems*, on Theoretical Chemistry symposium organized by School of Chemistry, University of Hyderabad. (14-17 Dec. 2016).
  - Abhisek Ghosal and Amlan K Roy, *DFT of many electron systems in Cartesian grid*, on Departmental Day of Chemical Sciences, IISER Kolkata. (Dec. 2015).
  - Abhisek Ghosal and Amlan K Roy, *DFT of atoms and molecules in Cartesian grid*, on Asian academic seminar and school organized by IISER Kolkata, IACS Kolkata, NINS and IMS of Japan. (6-10 March, 2015).
- **School and Workshop**
  - Scikit-Learn Workshop, Northwestern University IT, USA. (10-11 July, 2025)
  - Intermediate Python Workshop, Northwestern University IT, USA. (7-9 July, 2025)
  - Python Fundamentals, Northwestern University IT, USA. (30th June - 3rd July, 2025)
  - 2019 TDDFT School & Workshop: Excited states and dynamics, Rutgers University-Newark, USA. (5-14 August, 2019)
  - Computational electronic structure methods for atomic, molecular and solid-state systems, GIAN course at IIT-Bhubaneswar, India. (8-18 July, 2016)

Software Development

FHI-aims Contributions
All-electron electronic structure theory with numeric atom-centered orbitals (2026– Continue).

InDFT (Lead Developer)
A DFT program for atoms and molecules in CCG (Theoretical Chemistry Laboratory, IISER Kolkata, India, 2019) (2014– Continue).

TURBOMOLE Contributions
Program Package for Electronic Structure Calculations (2021-2023).

Publications

Published works

1. **Abhisek Ghosal** and George C. Schatz, Autoionized excited states of N2 molecule using complex-basis function spin-flip coupled cluster theory, *J. Chem. Phys*, **164**, 014118-1 (2026).

2. Ritaj Tyagi, **Abhisek Ghosal** and Vamsee K. Voora, Optical Absorption spectra of F-Center Defects in LiF using single-pole exchange-correlation and periodic electrostatic embedding, *Phys. Chem. Chem. Phys*, **27**, 18567-18576 (2025).

3. Nikhil S Chellam, Subhajyoti Chaudhuri, **Abhisek Ghosal**, Sajal K Giri and George C. Schatz, Density Functional Tight-Binding Enables Tractable Studies of Quantum Plasmonics, *J. Phys. Chem. C.*, **129**, 13839-13859 (2025).

4. **Abhisek Ghosal**, Pulkit Joshi and Vamsee K Voora, Taming Negative Ion Resonances Using Nonlocal Exchange-Correlation Functionals, *J. Phys. Chem. Lett.*, **15**, 5994-6001, (2024).

5. **Abhisek Ghosal** and Amlan K. Roy, A real-time TDDFT scheme for strong-field interaction in Cartesian coordinate grid, *Chem. Phys. Lett.*, **796**, 139562, (2022).

6. Raj Roy, **Abhisek Ghosal** and Amlan K. Roy, Charge-Transfer Excitation within a Hybrid-(G) KS Framework through Cartesian Grid DFT, *J. Phys. Chem. A*, **126**, 1448, (2022).

7. **Abhisek Ghosal** and Amlan K. Roy, A self-consistent systematic optimisation of range-separated hybrid functionals from *first principles*, *Mol. Phys.* **e1983056** (2021).

8. Raj Roy, **Abhisek Ghosal** and Amlan K Roy, A Simple Effective SCF Method for Computing Optical Gaps in Organic Chromophores, *Chemistry–An Asian Journal* **16** 2729 (2021).

9. **Abhisek Ghosal**, Tarun Gupta, Kishalay Mahato and Amlan K Roy, The effectiveness of pseudopotential Cartesian-grid KS-DFT on single excitation energies through Becke's exciton model, *Theor. Chem. Acc.* **140** 1 (2021).

10. Jan Gerit Brandenburg, Kieron Burke, Jannis Erhard, Emmanuel Fromager, **Abhisek Ghosal** and *et al.*, New density-functional approximations and beyond: general discussion, *Faraday Discussions* **224** 166 (2020).

11. **Abhisek Ghosal**, Tanmay Mandal and Amlan K. Roy, Efficient HF exchange evaluation through Fourier convolution in Cartesian grid for orbital-dependent density functionals, *J. Chem. Phys.* **150** 064104-9 (2019).

12. Amlan K. Roy, **Abhisek Ghosal** and Tanmay Mondal, InDFT: A DFT Program for Atoms and Molecules in CCG, IISER Kolkata, India: Theoretical Chemistry Laboratory, (2019).

13. Tanmay Mandal, **Abhisek Ghosal** and Amlan K. Roy, Static polarizability and hyperpolarizability in atoms and molecules through a Cartesian-grid DFT, *Theor. Chem. Acc.* **138** 10 (2019).

14. **Abhisek Ghosal**, Tanmay Mandal and Amlan K. Roy, Density functional electric response properties of molecules in Cartesian grid, *Int. J. Quant. Chem.* **118** e25708 (2018).

15. **Abhisek Ghosal**, Neetik Mukherjee and Amlan K. Roy, Information entropic measures of a quantum harmonic oscillator in symmetric and asymmetric confinement within an impenetrable box, *Ann. Phys.* (Berlin) **528** 796-818 (2016).

Book Chapters

1. **Abhisek Ghosal** and Amlan K. Roy, DFT calculations of atoms and molecules in Cartesian grid", in *Specialist Periodical Reports: Chemical Modelling, Applications and Theory*, **13** 221-260, Eds. M. Springborg and J.-O.Joswig, Royal Society of Chemistry, London (2016).

Submitted

1. Ken Miyazaki* and **Abhisek Ghosal***, Vibronic anatomy of reverse intersystem crossing through singularity-free second order nonradiative rate formulation, *submitted* (2026).

2. Raj Roy and **Abhisek Ghosal***, Simple model for challenging excited states : low-lying, core-level and conical intersection regimes, *submitted* (2026).

Works in Progress

1. **Abhisek Ghosal**, Julio C. V. Chagas and George C. Schatz, Performance of complex absorbing potential based multireference configuration interaction methods for autoionizing excited states of N_2, *Manuscript under preparation.*

2. **Abhisek Ghosal**, Raj Roy and George C. Schatz, Probing Resonant Inelastic X-ray Spectrum through a modified ΔSCF-based exciton model, *Manuscript under preparation.*

References

- George C. Schatz
  Morrison Professor of Chemistry; Professor of Chemical and Biological Engineering
  Department of Chemistry
  Northwestern University, 2145 Sheridan Rd., Evanston IL 60208-3113 USA
  (847)491-5657 (voice) (847) 491-7713 (fax) email: g-schatz@northwestern.edu
  Office location: Room 4017, Ryan Hall, 2190 Campus Dr., Northwestern University
- Prof. Amlan K. Roy
  Associate prof., Department of Chemical sciences
  IISER Kolkata, Mohanpur-741246, WB, India
  E-mail: akroy@iiserkol.ac.in, akroy6k@gmai.com
  Webpage: https://www.iiserkol.ac.in/ theochem/index.html
- Dr. Vamsee K. Voora
  Department of Chemistry
  TIFR Mumbai, 400005, Maharastra
  Phone: +91 22 2278 2978
  E-mail: vamsee.voora@tifr.res.in
  Webpage : https://vkvoora.github.io
- Dr. Attila Cangi
  Head of Department
  Machine Learning for Materials Design
  Helmholtz-Zentrum Dresden-Rossendorf
- Dr. Niranjan Govind
  Environmental Molecular Sciences Laboratory
  Pacific Northwest National Laboratory
  E-mail: niri.govind@gmail.com
- Prof. Neetu Goel
  Department of Chemistry and Center of Advanced Studies in Chemistry
  Panjab University, Chandigarh-160014, India
  Phone: +91 9815970112
  E-mail: neetugoel@pu.ac.in